Portfolio Performance

Portfolio Performance Measures

- Sharpe’s Measure
- Treynor’s Measure
- Jensen’s Measure
- Fama’s Three Factor Model
- Carhart’s Four Factor Model
- Portfolio Style Evaluation
- Portfolio Attribute Analysis
- Portfolio Tracking Error
- Portfolio Turnover

Sharpe's Measure

$$\frac{\text{Expected Excess Return}}{\sigma_{\text{port}}}$$

$$\frac{E(R_p - RFR)}{\sigma_{\text{port}}}$$

- Where
  - *E(Rp – RFR)* refers to the expected excess returns of the portfolio (Rp) over the risk free return (RFR) rate.
- Slope is computed by dividing this expected excess return by the standard deviation of the portfolio.

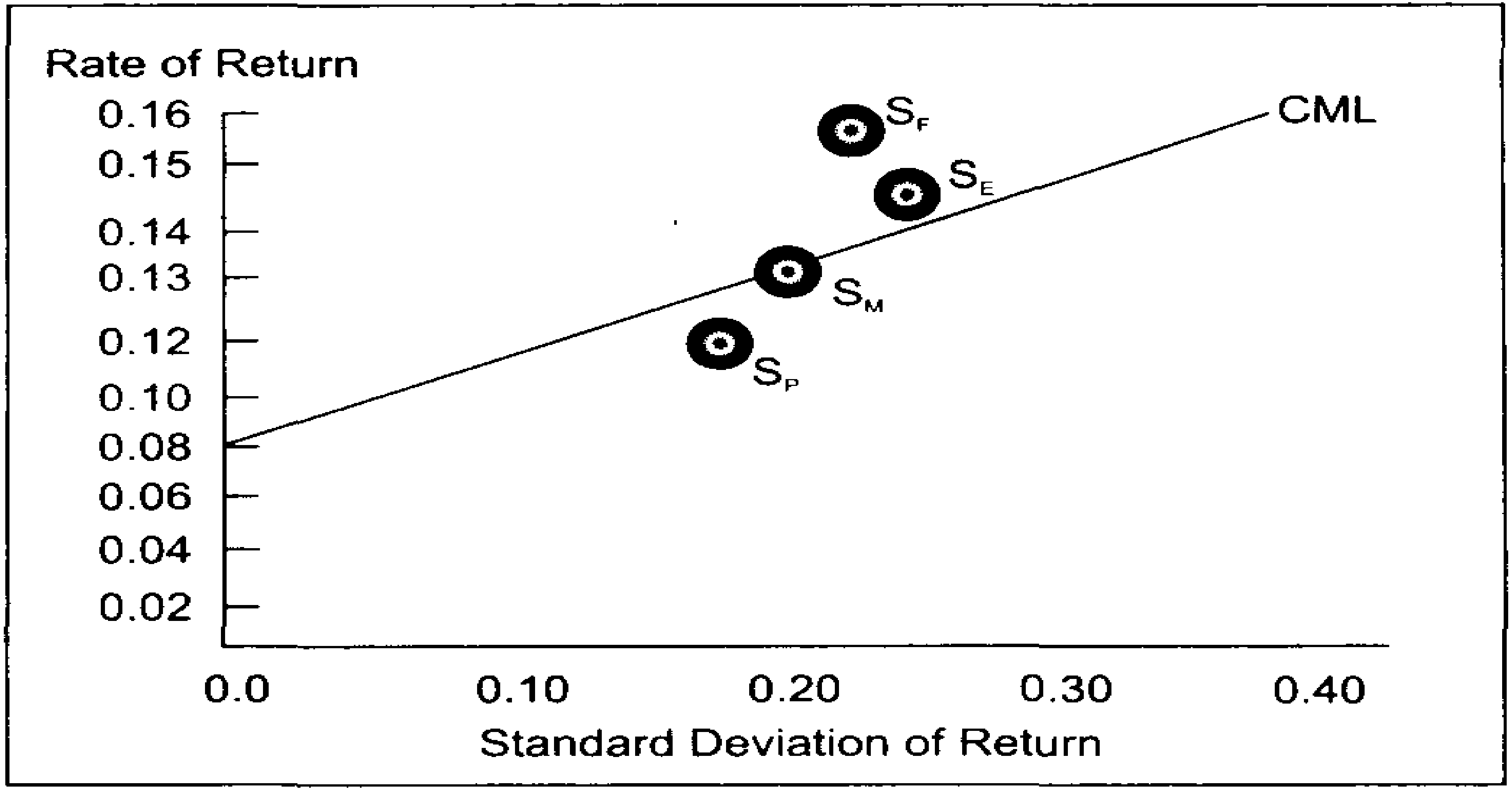
Sharpe’s Measure (Performance Evaluation Measure)

$$\boxed{S = \frac{\left(R_{port} - RER\right)}{\sigma_{port}}}$$

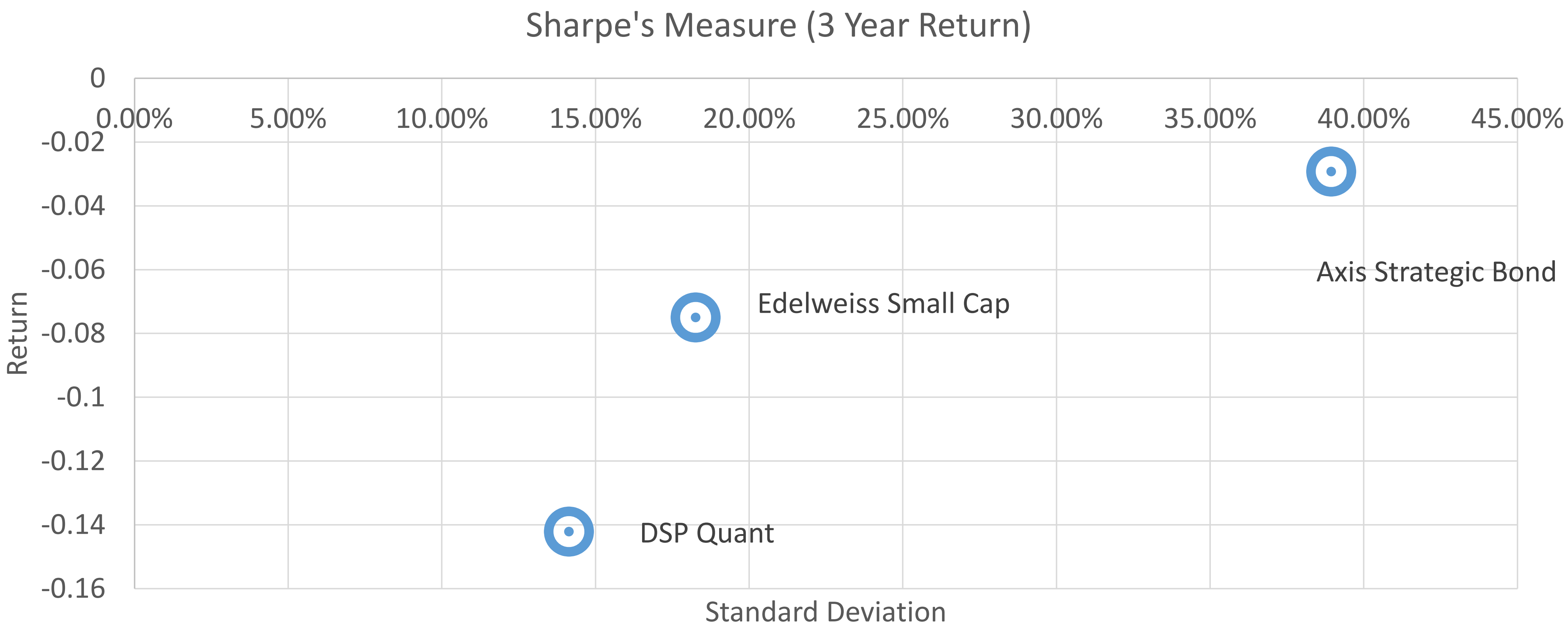
- Where:
  - ‘S’ indicates Sharpe's measure;
  - ‘RFR’ refers to risk free rate
  - ‘R_{port}’ refers to portfolio return
  - ‘σ_{port}’ refers to portfolio standard deviation.

- This composite measure of portfolio performance seeks to measure the total risk of the portfolio by including the standard deviation of returns rather than considering only the systematic risk by using beta.

Sharpe's Measure



Fund Performance (Sharpe’s Measure)



Sharpe's Measure

- Historic (ex post) Sharpe's ratio (S)

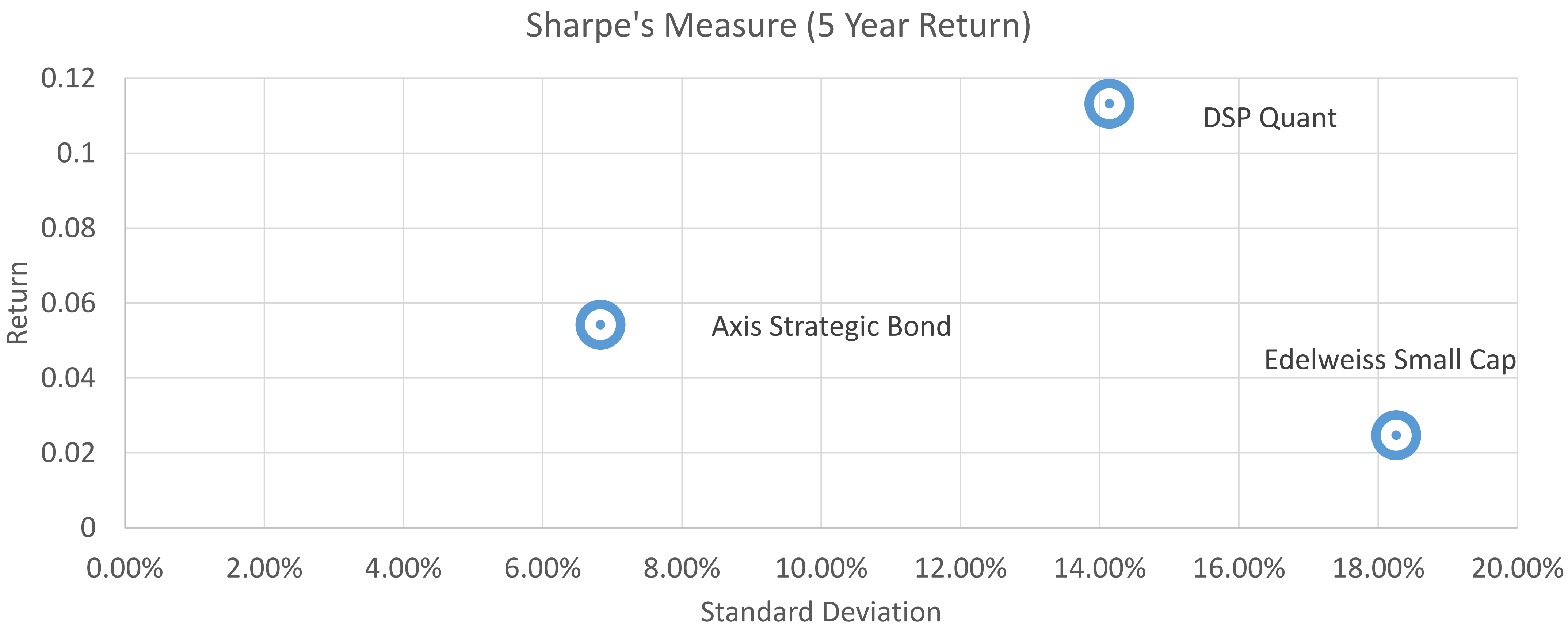
$$S = \frac{\bar{D}}{\sigma_p}$$

- where
  - R_{Pl} = the return on a portfolio in time t
  - R_{Bt} = the return on the benchmark portfolio in time t
  - D_t - the differential return in time t
  - $D_t = R_{Pt} - R_{Bt}$
  - *Average D* = the average value of D_t over the period being examined

$$\bar{D} = \frac{\sum_{t=1}^{n} D_t}{N}$$

  - σ_P = the standard deviation for the portfolio during the period

Fund Performance (Sharpe’s Measure)

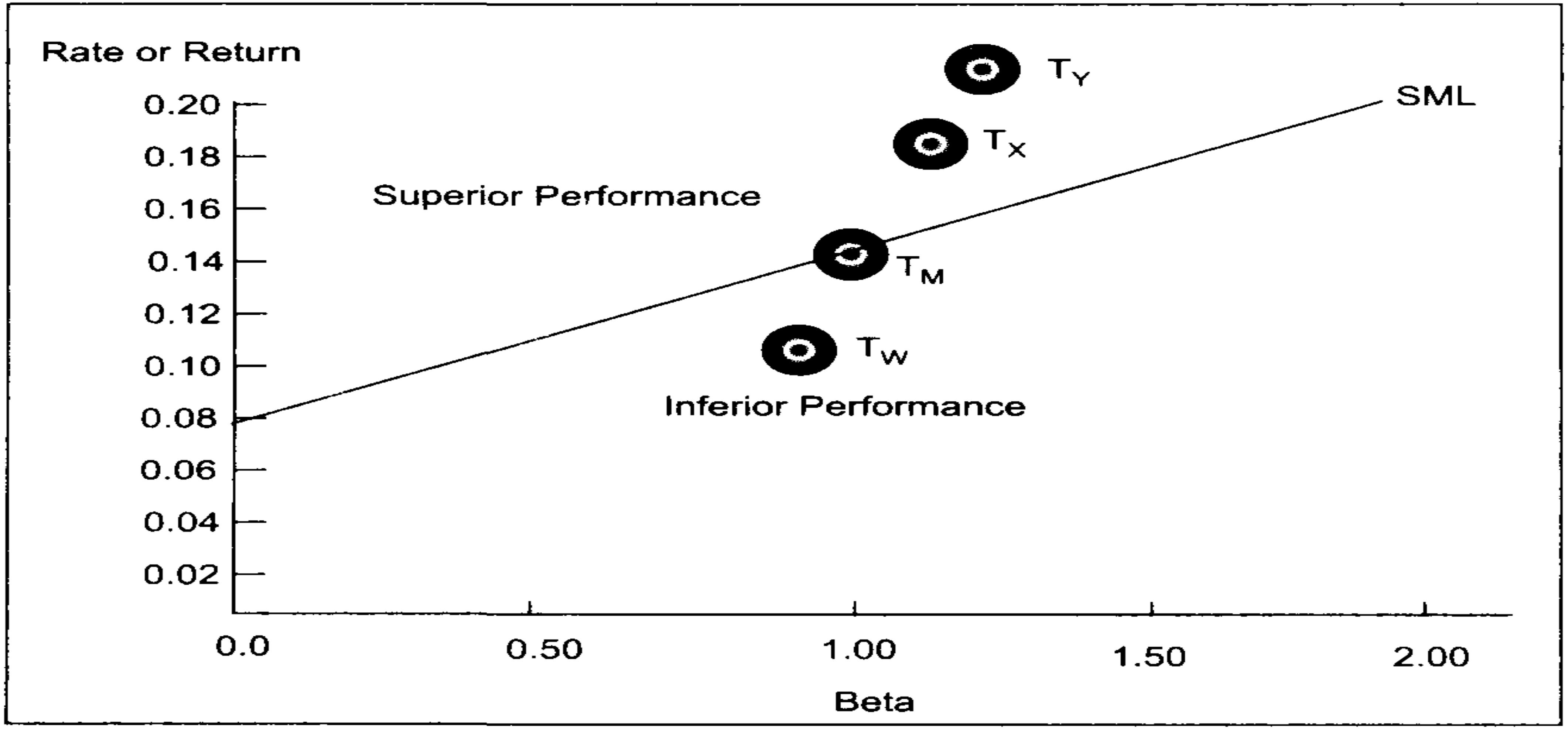


Treynor's Measure

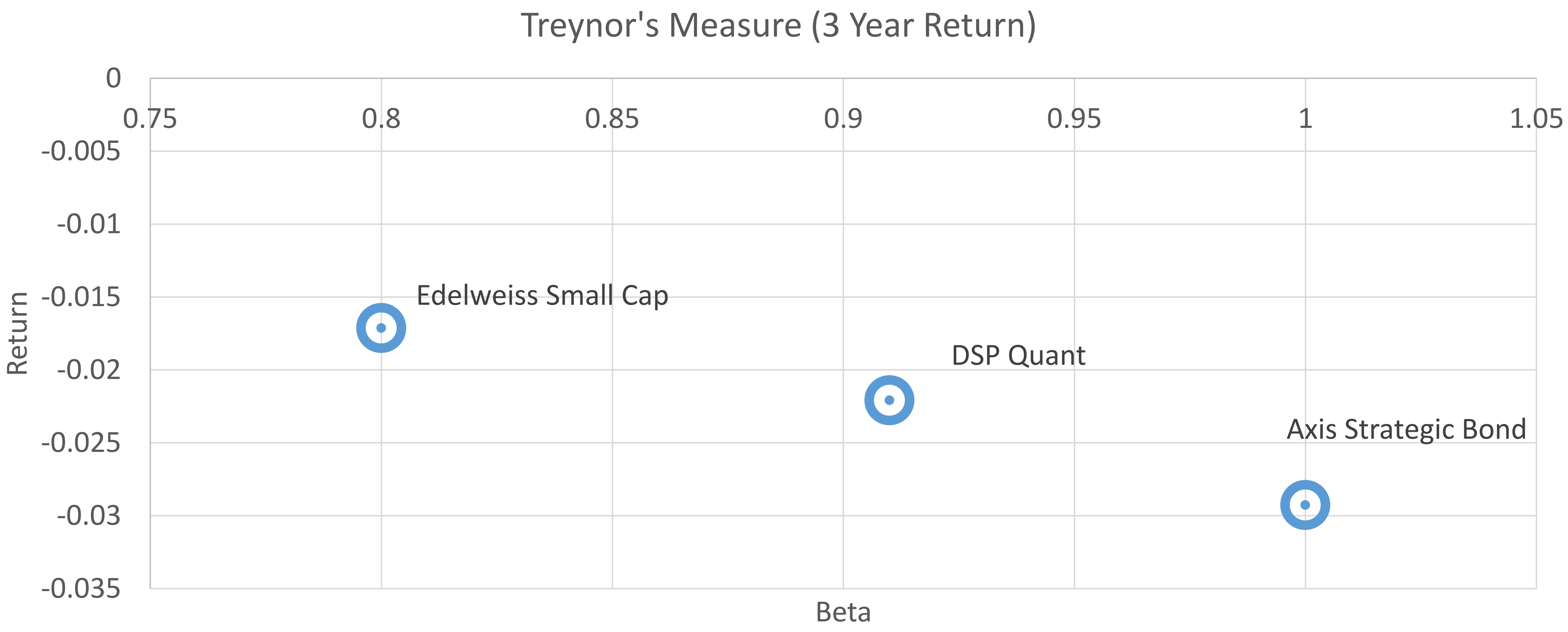
$$T = \frac{(R_{\text{port}} - RFR)}{\beta_{\text{port}}}$$

- Numerator of the ratio is the risk premium
- Denominator is a systematic measure of risk
- Measure indicates the portfolio's risk premium return per unit of systematic risk

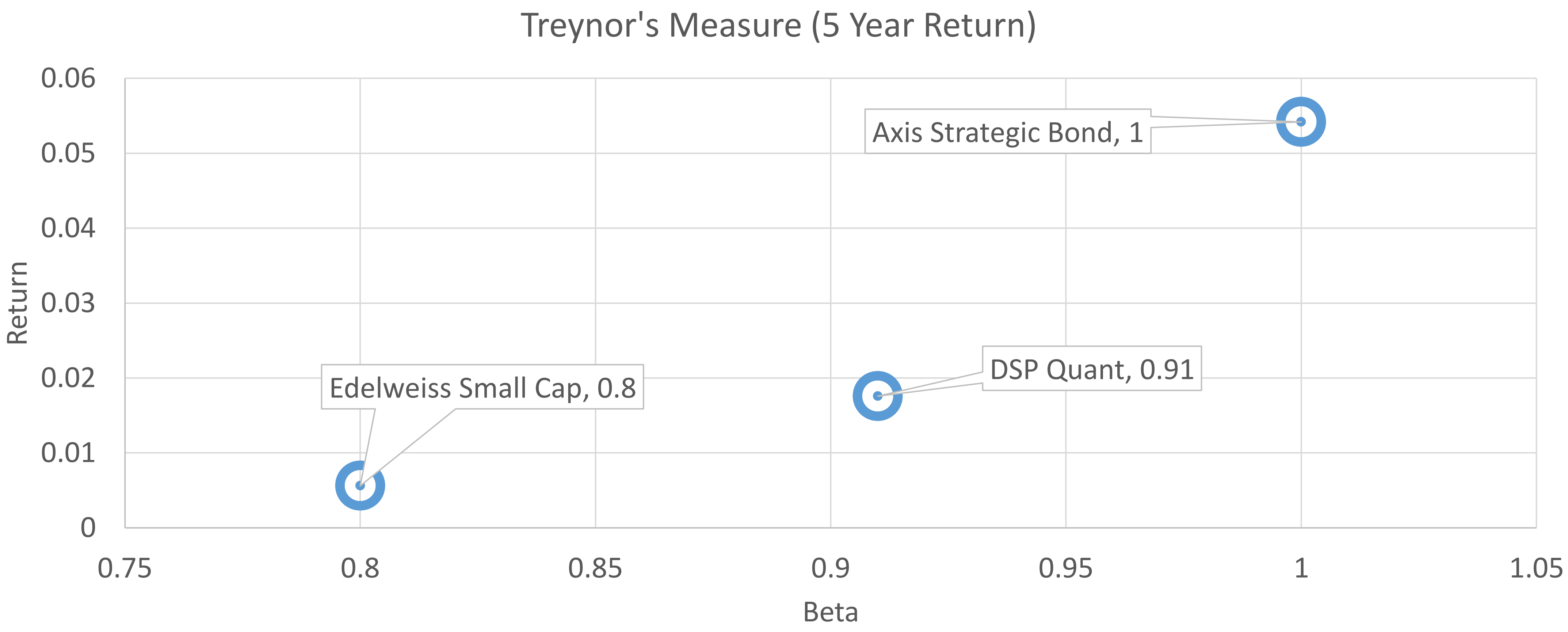
Treynor’s Measure



Fund Performance (Treynor's Measure)



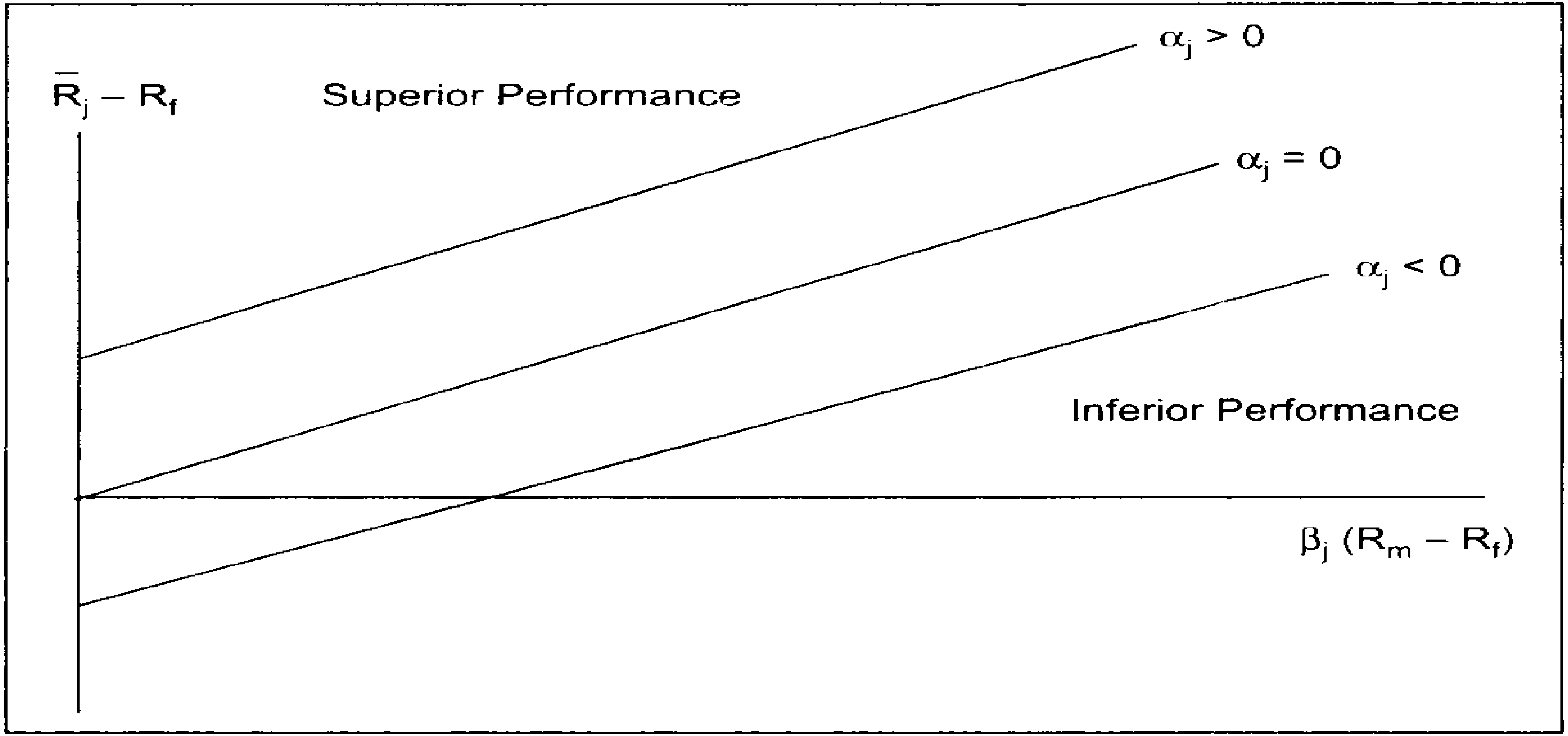
Fund Performance (Treynor's Measure)



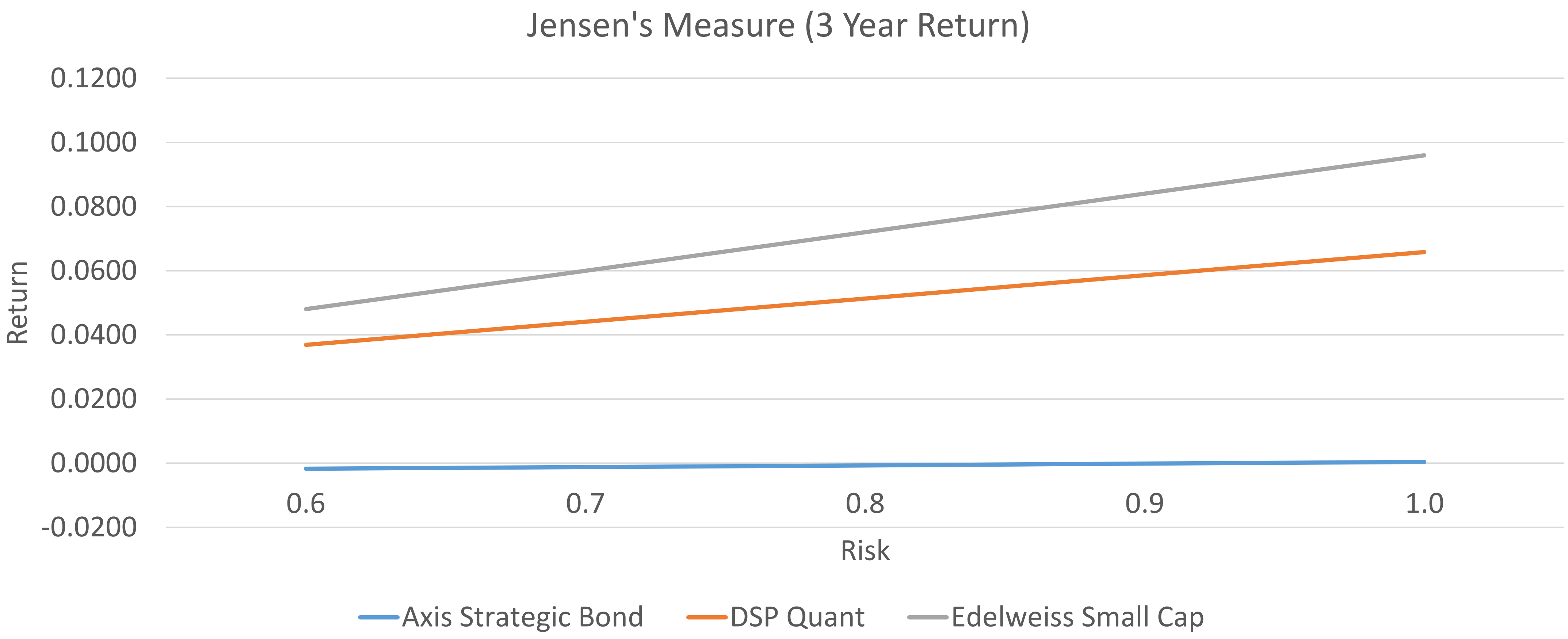
Jensen’s Measure

- $R_{jt} – RFR_t = \beta j[R_{mt} – RFR_t] + U_{jt}$
- Risk premium earned on portfolio *j* is equal to β times a market risk premium, plus a random error term

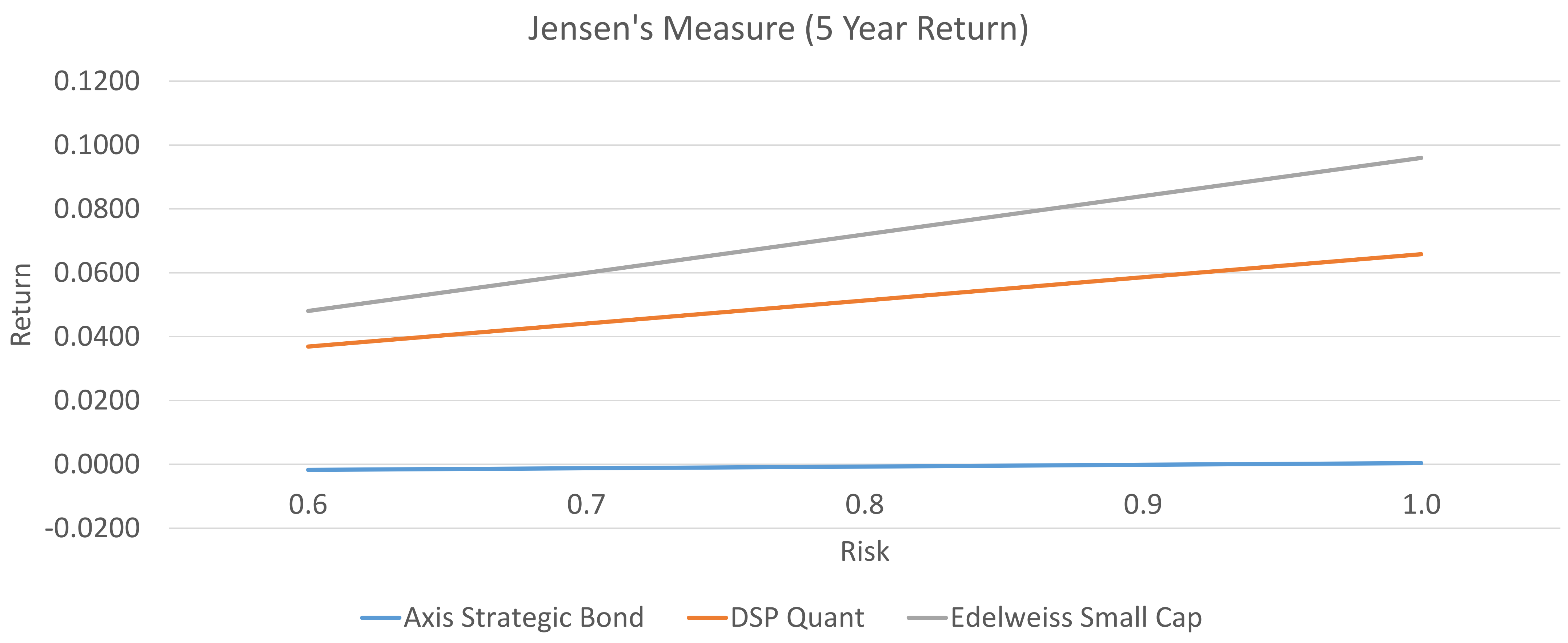
Jensen's Measure



Fund Performance (Jensen's Measure)



Fund Performance (Jensen’s Measure)



Fama’s Three Factor Model

- Fama’s three factor model considers market risk premium (Rm-Rf), size risk (SMB) and value risk (HML).

$$R_i = \alpha_i + \beta_i \left(R_m - R_f \right) + \delta_i SMB + \lambda_i HML + \varepsilon_i$$

R_i – Portfolio return; R_m – Market return; R_f – Risk free return;

α_i –Portfolio alpha; β_i – Portfolio beta; δ_i – Size coefficient;

λ_i – Value coefficient; ε_i – Error term.

Fama’s Three Factor Model

- Size risk measured as SMB (small value stocks minus big value stocks) indicates the size difference between big and small firms

$$SMB_t = \frac{\sum_{i=1}^{n} R_{Sit}}{n} - \frac{\sum_{i=1}^{m} R_{Bit}}{m}$$

SMB_t – Size mimicking portfolio return at time 't'

R_{Sit} – Return of small size firm 'i' at time 't'

R_{Bit} – Return of big size firm 'i' at time 't'

n – Number of small size firms in benchmark portfolio (30% of small size firms in benchmark portfolio)

m – Number of big size firms in benchmark portfolio (30% of big size firms in benchmark portfolio)

Fama’s Three Factor Model

- Value risk has been defined to measure the value premium provided by portfolios through investment in stocks that have high book value to market value ratio

$$HML_t = \frac{\sum_{i=1}^{n} BTM_{Hnt}}{n} - \frac{\sum_{i=1}^{m} BTM_{Lmt}}{m}$$

HML_t – Value factor of the Three factor model at time 't'

BTM_{Hnt} – Value of high book to market value firms

(50% of high book to market value firms in benchmark portfolio) at time 't'

BTM_{Lmt} – Value of low book to market value firms

(50% of low book to market value firms in benchmark portfolio) at time 't'

n – Number of high book to market value firms; m – Number of low book to market value firms

Carhart's Four Factor Model

- Carhart's four factor model included portfolio momentum risk besides size and value risk that portfolio managers use to enhance portfolio returns.
- Lagged momentum factor can explain extraordinary returns of a portfolio.
- Carhart's model includes size, value and momentum risk to explain total portfolio returns.

$$\boxed{\begin{array}{l} R_i = \alpha_i + \beta_i \left(R_m - R_f \right) + \delta_i SMB + \lambda_i HML + \rho_i MOM + \varepsilon_i \\ R_i - \text{Portfolio return; } R_m - \text{Market return; } R_f - \text{Risk free return;} \\ \alpha_i - \text{Portfolio alpha; } \beta_i - \text{Portfolio beta; } \delta_i - \text{Size coefficient;} \\ \lambda_i - \text{Value coefficient; } \rho_i - \text{Momentum coefficient; } \varepsilon_i - \text{Error term.} \end{array}}$$

Carhart's Four Factor Model

- Momentum Risk Factor

$$MOM_t = \frac{\sum_{i=1}^{n} R_{Hit}}{n} - \frac{\sum_{i=1}^{m} R_{Lit}}{m}$$

MOM_t – Lagged momentum mimicking portfolio return at time 't'

R_{Hit} – Return of highest return security 'i' at time 't'

R_{Lit} – Return of lowest return security 'i' at time 't'

n – Number of highest return securities firms in benchmark portfolio (30% of highest return securities in benchmark portfolio)

m – Number of lowest return securities in benchmark portfolio (30% of lowest return securities in benchmark portfolio)

Four Factor Model Fund Performance

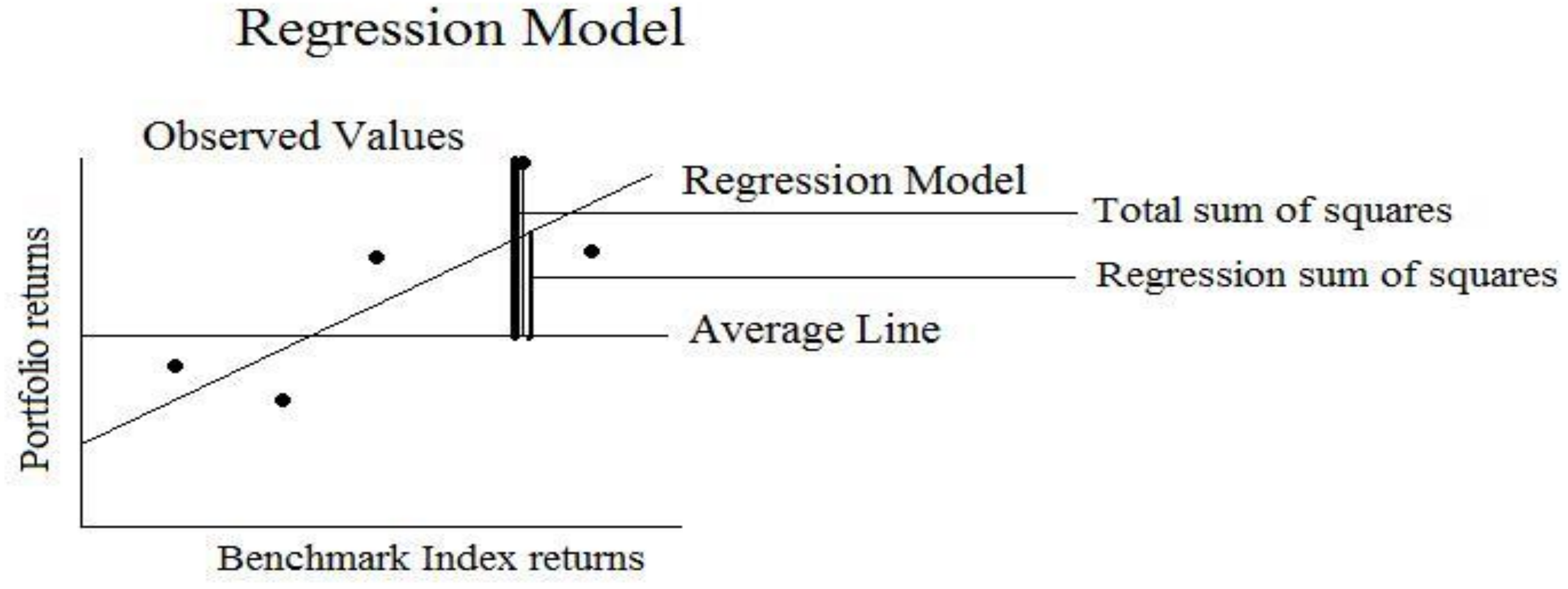
Regression Statistics	
Multiple R	0.967229
R Square	0.935531
Adjusted R Square	0.921205
Standard Error	0.002482
Observations	23

	df	*SS*	*MS*	*F*	*Significance F*
Regression	4	0.001609	0.000402	65.30141	1.81E-10
Residual	18	0.000111	6.16E-06		
Total	22	0.00172			

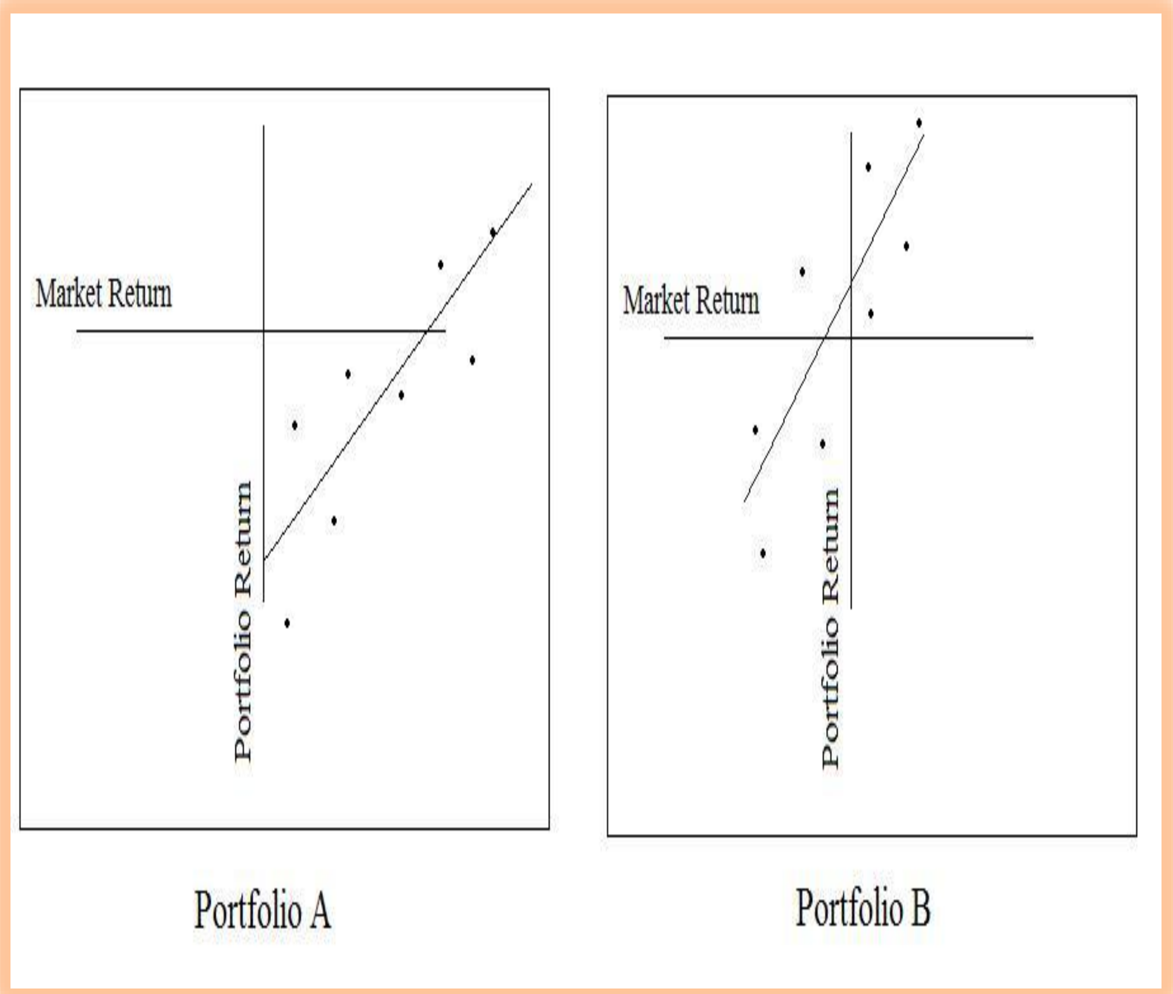
Fund Manager Performance: Edelweiss

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	0.105593	0.08558	1.233855	0.233126
Nifty small cap 250	0.975283	0.077023	12.66222	2.12E-10
Size	0.002509	0.001213	2.069073	0.05321
Value	-0.01138	0.008783	-1.29534	0.211572
MOM	0.001252	0.001078	1.161618	0.260563

Portfolio Style Evaluation



Portfolio Style Evaluation



- Portfolio A has a slope that is lower than slope B, hence portfolio manager B could be said to have **timed** the market better.
- Negative intercept of portfolio A indicates that the **security selection** ability of portfolio manager A is not good.
- Portfolio manager B has recorded a positive slope indicating better security selection skill.

Portfolio Attribute Analysis

Portfolio Sector Allocation						
	Benchmark Weights	**Benchmark Return**	**Expected Return**	**Actual Weights**	**Benchmark Return**	**Reported Return**
Share	60%	15%	9.0	70%	15%	10.5
Bond	40%	10%	4.0	30%	10%	3.0
			13.0%			13.5%

Portfolio Security Selection						
Policy	**Actual Weights**	**Benchmark Return**	**Expected Return**	**Actual Weights**	**Actual Return**	**Reported Return**
Share	70%	15%	10.5	70%	14%	9.8
Bond	30%	10%	3.0	30%	12%	3.6
			13.5%			13.4%

Portfolio Attribute Analysis

- Total Portfolio Return
- Reported return – Expected return = +0.40%

Policy	Benchmark Weights	Benchmark Return	Expected Return	Actual Weights	Actual Return	Reported Return
Share	60%	15%	9.00%	70%	14%	9.80%
Bond	40%	10%	4.00%	30%	12%	3.60%
			13.00%			13.40%

Portfolio Attribute Analysis

Equity Portfolio

	Portfolio		Benchmark	
Component	Weight	Return	Weight	Return
Gas and petrochemical products	6.05	22.32	8	20.57
Banks	10.14	23.92	10	20.36
Software and Consultancy Services	6.97	26.48	6	25.95
Auto & Auto Ancillaries	2.52	15.98	4	18.53
FMCG	2.48	24.76	4	25.78
Pharmaceuticals & Biotechnology	2.28	12.15	4	17.43
Telecommunications Equipment Manufacturer	2.28	12.91	4	10.45
Total	32.72	22.03	40	20.32

Debt Portfolio

	Portfolio		Benchmark	
Component	Weight	Return	Weight	Return
Sovereign instruments	15.59	6.45	15	4.67
Financial Institutions	41.34	8.35	35	5.92
Corporate	10.35	12.56	10	9.86
Total	67.28	8.56	60	6.26

Portfolio Sector Attribute Analysis

Equity Portfolio

Component	Allocation	Selection
Gas and petrochemical products	-0.31	0.32
Banks	1.22	1.10
Software and Consultancy Services	1.64	0.11
Auto & Auto Ancillaries	-0.43	-0.20
FMCG	-0.62	-0.08
Pharmaceuticals & Biotechnology	-0.53	-0.37
Telecommunications Equipment Manufacturer	-0.32	0.17
Total	0.65	1.07

Debt Portfolio

Debt	Allocation	Selection
Component	-0.09	0.41
Sovereign instruments	0.18	1.49
Financial Institutions	-0.13	0.42
Corporate	-0.03	2.32
Total	-0.06	4.64

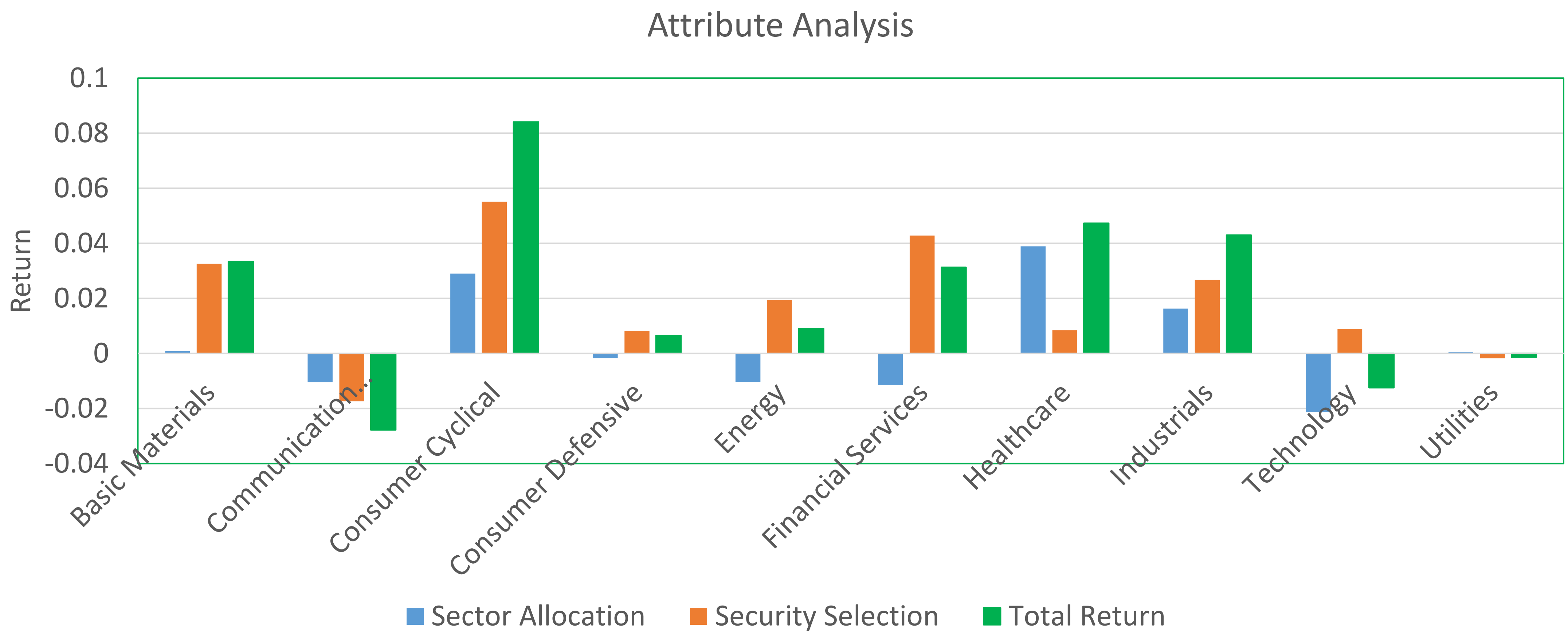
Portfolio Attribute Analysis

	Allocation	Selection	Total
Equity	-1.48	0.56	
Debt	0.46	1.54	
Total	-1.02	2.11	1.08

Equity Fund Attribute Analysis: Edelweiss

	Benchmark		Edelweiss	
Sector	**Weight**	**Return**	**Weight**	**Return**
Basic Materials	0.0513	0.150717	0.0570	0.72104
Communication Services	0.0450	0.793654	0.0319	0.247321
Consumer Cyclical	0.0842	0.526093	0.1393	0.921798
Consumer Defensive	0.0849	0.149487	0.0735	0.26174
Energy	0.1082	0.255855	0.0677	0.54327
Financial Services	0.3604	0.147891	0.2828	0.299193
Healthcare	0.0208	0.639697	0.0816	0.742413
Industrials	0.0459	0.343933	0.0932	0.630702
Technology	0.1621	0.317812	0.0949	0.411407
Utilities	0.0373	0.726583	0.0379	0.679128

Equity Fund Attribute Analysis: Edelweiss



Portfolio Tracking Error

- Annualized standard deviation of the difference in returns between the portfolio and its target index.

- Mutual funds: Logarithmic returns on the NAV per unit and its benchmark for a reporting period (annual percentage).

Reasons for Tracking Error

- Trading costs
- Portfolio security weights vs. benchmark security weights
- Impact of placing the security trade in a portfolio
- Management time
- Portfolio tax implications
- Window dressing
- Reinvestment of periodic contributions to portfolio (dividend)

Fund Tracking Error

Date	Fund NAV A	Fund NAV B	Benchmark	Difference A	Difference B
01-Jul	279.814	66.514	17771.4		
04-Jul	281.289	66.7446	17838.05	0.001514	-0.00028
05-Jul	280.837	66.7587	17824.2	-0.00083	0.000988
06-Jul	282.542	67.0425	18058.6	-0.00701	-0.00882
07-Jul	284.993	67.0963	18264.3	-0.00269	-0.01052
08-Jul	286.777	67.1924	18347.85	0.001676	-0.00313
11-Jul	287.387	67.2082	18384.55	0.000127	-0.00176
12-Jul	286.105	67.1153	18206.2	0.005278	0.008365
13-Jul	285.174	67.1859	18193.45	-0.00256	0.001752
14-Jul	284.42	67.1316	18159.4	-0.00077	0.001065
15-Jul	285.434	67.1479	18293.05	-0.00377	-0.00709
18-Jul	288.174	67.3922	18550.1	-0.0044	-0.01032
19-Jul	289.705	67.4874	18612.1	0.001962	-0.00193
20-Jul	291.136	67.5818	18745.95	-0.00224	-0.00577
21-Jul	292.191	67.6811	18910.5	-0.00512	-0.00727
22-Jul	292.686	67.8076	19008.4	-0.00347	-0.0033
25-Jul	293.056	67.8057	18947	0.004499	0.003207
26-Jul	291.68	67.6637	18797.5	0.003215	0.005825
27-Jul	293.908	67.8105	18994.7	-0.00283	-0.00827
28-Jul	295.651	67.9925	19279.3	-0.00896	-0.01219
29-Jul	298.202	68.1981	19585.55	-0.00717	-0.01274
Annualized standard deviation				0.061973	0.095781

Tracking Error

Scheme	Benchmark	Tracking Error Regular (%)	Tracking Error Direct (%)
Aditya Birla Sun Life Nifty 50 Equal Weight Index Fund	NIFTY 50 Equal Weight Total Return Index	0.13	0.13
Aditya Birla Sun Life Nifty 50 ETF	NIFTY 50 Total Return Index	0.05	--
Aditya Birla Sun Life Nifty 50 Index Fund	NIFTY 50 Total Return Index	0.07	0.07
Aditya Birla Sun Life Nifty Bank ETF	NIFTY Bank Total Return Index	0.06	--
Aditya Birla Sun Life Nifty IT ETF *	NIFTY IT Total Return Index	0.09	--
Aditya Birla Sun Life Nifty Midcap 150 Index Fund	NIFTY Midcap 150 Total Return Index	0.14	0.14
Aditya Birla Sun Life Nifty Next 50 ETF	NIFTY Next 50 Total Return Index	0.05	--
Aditya Birla Sun Life Nifty Next 50 Index Fund *	NIFTY Next 50 Total Return Index	0.26	0.26
Aditya Birla Sun Life Nifty Smallcap 50 Index Fund	NIFTY Smallcap 50 Total Return Index	0.14	0.14

Source: https://www.amfiindia.com/research-information/other-data/tracking_errordata

Tracking Error

Scheme	Benchmark	Tracking Error Regular (%)	Tracking Error Direct (%)
Aditya Birla Sun Life Nifty 200 Momentum 30 ETF *	Nifty 200 Momentum 30 Total Return Index	0.06	--
Aditya Birla Sun Life Nifty 200 Quality 30 ETF *	NIFTY 200 Quality 30 Total Return Index	0.07	--
Aditya Birla Sun Life Nifty 50 Equal Weight Index Fund	NIFTY 50 Equal Weight Total Return Index	0.13	0.13
Aditya Birla Sun Life Nifty 50 ETF	NIFTY 50 Total Return Index	0.05	--
Aditya Birla Sun Life Nifty 50 Index Fund	NIFTY 50 Total Return Index	0.07	0.07
Aditya Birla Sun Life Nifty Bank ETF	NIFTY Bank Total Return Index	0.06	--

Source: https://www.amfiindia.com/research-information/other-data/tracking_errordata

Tracking Error

Scheme	Benchmark	Tracking Error Regular (%)	Tracking Error Direct (%)
Aditya Birla Sun Life Nifty Healthcare ETF *	NIFTY Healthcare Total Return Index.	0.02	--
Aditya Birla Sun Life Nifty IT ETF *	NIFTY IT Total Return Index	0.09	--
Aditya Birla Sun Life Nifty Midcap 150 Index Fund	NIFTY Midcap 150 Total Return Index	0.14	0.14
Aditya Birla Sun Life Nifty Next 50 ETF	NIFTY Next 50 Total Return Index	0.05	--
Aditya Birla Sun Life Nifty Next 50 Index Fund *	NIFTY Next 50 Total Return Index	0.26	0.26
Aditya Birla Sun Life Nifty Smallcap 50 Index Fund	NIFTY Smallcap 50 Total Return Index	0.14	0.14

Source: https://www.amfiindia.com/research-information/other-data/tracking_errordata

Portfolio Turnover

- Lower of the total of new stocks purchased or sold over 12 months divided by the fund's average assets under management.

$$\frac{Min(S_b,S_s)}{\sum_1^n AUM/_n}$$

- S_b = Securities bought; S_s=Securities sold, AUM = Asset under management; n = Number of trades

Portfolio Turnover

Scheme Name	Funds Mobilized for the month of July (INR in crore)	Repurchase/ Redemption for the month of July (INR in crore)	Average Net Assets Under Management for the month July (INR in crore)	Portfolio Turnover
Hybrid Schemes				
Conservative Hybrid Fund	353.19	333.38	21,483.82	0.02
Balanced Hybrid Fund/Aggressive Hybrid Fund	2,166.56	1,480.38	1,45,724.53	0.01
Dynamic Asset Allocation/Balanced Advantage Fund	2,692.38	2,138.44	1,83,463.51	0.01
Multi Asset Allocation Fund	388.25	210.49	20,825.41	0.01
Arbitrage Fund	3,552.06	9,959.61	1,06,812.63	0.03
Equity Savings Fund	388.47	564.95	17,592.83	0.02
Sub Total - III (i+ii+iii+iv+v+vi)	9,541	14,687	4,95,903	

Data source: https://www.amfiindia.com/research-information/amfi-monthly